VIRTUAL SCHOOL VICTORIA

Algorithmics Unit 3

School Assessed Task 2026 – Mining For Uranium

Scenario

Energy Australia is surveying Victoria for uranium for a nuclear power plant. You are assisting the geological survey team to design an algorithm to find an optimal route to travel to mine this precious resource.

Each location in the state contains a relatively small quantity of uranium, and it takes different amounts of time to mine at different locations due to various local factors. For example, it might be the case that in Shepparton there are 30 units of uranium available and it will take 3 hours to extract; at the same time, in Horsham there are 25 units available, but it will take 4 hours to extract.

Your team is only allowed to work within a given radius of a given location, and only for a limited time.

For example, you might receive a permit to work within 300km of Ballarat for up to 36 hours. You must find the best route which

- stays inside the given radius
- completes both travel and extraction within the given time
- can start and finish at any place within the radius and does not need to visit every location

The algorithm you design may later be deployed to larger situations with similar needs.

Report

For Unit 3, you are completing Part 1 of the overall SAT, where you will design a model and algorithm to solve the problem as stated.

Your report should contain the following sections:

1. Model and Specification

- Clearly specify the problem, including discussion of the relevant real-life information you intend to model.
- State which ADTs you will use to model the problem and justify your choice.
- Provide signatures for the operations for each ADT.

Depending on how you model the problem, you will need to create additional ADTs. You will also need to make some decisions about how you interpret the problem. These should be explained fully in order to achieve the highest levels.

2. Design your Algorithm

- Explain why you chose a particular approach, comparing features of your algorithm to other options.
- Clearly explain the broad overview/approach of each algorithm.

3. Code your Algorithm

- Write your algorithm in pseudocode. Include sufficient comments and explanations to make your intentions clear.

4. Evaluate your model and algorithm

- Evaluate/justify how well your model is suited to the real life situation and how well it meets its purpose. (You do not need to get into any Unit 4 style time complexity analysis at this stage)

Authentication

- This is an individual project. You can discuss it with others, but the work must be your own.
- Students must be able to explain their work if called upon.
- You must acknowledge any external resources you draw upon.
- Your score will rank you within the cohort, so it is not in your interests to collude.
- You must not submit any work that has been generated using artificial intelligence, LLM etc. If such work is detected you may be awarded zero.

Links to Unit 4

In Unit 4, you will be given a complete solution to this problem written in Python. In Part 2, you will analyse this algorithm for time complexity. Finally, in Part 3, you will use advanced algorithm design features to improve it. Such features include:

- dynamic programming
- backtracking
- hill climbing
- simulated annealing
- A*

You should avoid using these features in Part 1 of the SAT, and you are advised to stick to designs you have seen in Unit 3.

Other tips

- Keep an eye on the news forum for further advice that will undoubtedly arise as students work on the project.
- Start work early so you can ponder different approaches and later check your code for errors.

Timeline

Your final report is due on 29th May 2026. It should be submitted via VSV online.

At the end of your work, include the sentence "I declare that this is my own work and has been completed according to the specified conditions."

You can ask clarifying questions if you are unsure how to proceed. Answers will generally be shared with the cohort on the news forum on VSV online.

You must submit a 'draft' of whatever you have done **by 1st May 2026** on VSV Online. This is to ensure that you are on track to complete your work, and for authentication purposes. It will not be formally marked, however it will be used to inform feedback provided to the whole cohort.

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Assessment Criteria

Criterion 1: specify a problem and model its key features. Precisely specify the algorithm problem you intend to solve.

Criterion 2: design an algorithm to solve your problem, considering multiple approaches and combinations of algorithm patterns.

Criterion 3: communicate your algorithm using pseudocode.

Criterion 4: evaluate your model, considering the suitability of your algorithm, how it integrates with your model and whether or not it solves your original problem. (Formal analysis will come later in Unit 4).

You will receive a mark out of ten for each criterion, based on the following rubric.

		1-2 (very low)	3-4 (low)	5-6 (medium)	7-8 (high)	9-10 (very high)
1	Specifies an algorithmic problem.	Identifies some algorithmic aspects of the real world problem.	Outlines/explains some algorithmic aspects of the real world problem.	Formulates an algorithmic problem from the real world problem.	Explains how the algorithmic problem is formulated from the real world problem.	Provides a clear and precise specification of the algorithmic problem, formulated from the real world problem
1	Explains the salient features of the real world problem.	Lists arbitrary features of the real world problem.	Identifies some relevant features of the real world problem, giving reasons.	Describes salient features of the real world problem, giving reasons.	Chooses a suitable set of features to model based on consideration of the features of the real world model.	Chooses a **comprehensive** set of features to model based on consideration of the features of the real world model
1	Models the problem using ADTs.	Makes a limited attempt, ADTs may not be suitable.	Models some features using suitable ADTs.	Models selected features using a combination of suitable ADTs. Outlines how the problem maps to this model.	Models selected features using a **coherent** combination of suitable ADTs. Describes how the problem maps to this model. **Some signatures** are provided.	Models selected features using a **coherent and fit-for-purpose** combination of suitable ADTs. Describes how the problem maps to this model. **All signatures** for key operations are provided.

		1-2 (very low)	3-4 (low)	5-6 (medium)	7-8 (high)	9-10 (very high)
2	Considers suitable algorithmic approaches to the problem	Identifies an algorithm with some relevant to the problem	Outlines a few algorithm design approaches that could form the basis of a solution.	Considers relevant characteristics of several algorithm design approaches.	Compares the suitability of some algorithm design approaches to determine an appropriate approach.	**Thoroughly** compares the suitability of algorithm design approaches to determine the best approach.
2	Describes the design of an algorithmic solution to the real world problem	Identifies some aspects of an algorithm to solve a real world problem.	Outlines a simple solution to the problem.	Describes a non-trivial algorithm that some some aspects of the specified problem.	**Describes** an algorithm that solves the specified problem using some combination of algorithms or algorithm design patterns.	**Clearly explains** an algorithm that solves the specified problem using some combination of algorithms or algorithm design patterns.

		1-2 (very low)	3-4 (low)	5-6 (medium)	7-8 (high)	9-10 (very high)
3	Communicates the algorithmic solution in pseudocode	Limited elements are expressed in pseudocode. Some correct initialisation of variables and data structures.	Elements of the structure are expressed in pseudocode. Correct use of simple iteration and conditional control structures where appropriate.	Algorithm is expressed in pseudocode so that the structure of the design is clear. Includes nested iteration and recursion where appropriate.	Algorithm is expressed in pseudocode that mostly reflects the solution design. Errors are minor and do not affect the overall structure. Some attempt to use functional abstractions.	Algorithm is expressed in pseudocode correctly and precisely. A modular approach is used including ADTs and functional abstractions.
4	Justifies the solution to the real world problem.	Identifies relevant reasons in support of a solution.	Outlines the rationale for a solution based on limited consideration of pros and cons.	Justifies choice of solution based on comparative advantages over other approaches.	Justifies choice of solution based on arguments relating to its suitability, coherence or fitness for purpose.	Justifies choice of solution by clearly demonstrating its suitability, coherence or fitness for purpose.

Frequently Asked Questions

Q. Can I get feedback on my work before I submit?

No. In order to provide the same level of support to all students, you are provided with all of the information you need in order to approach the task. You can see previous examples of highly performing work to help you to see what is expected.

Q. Can I ask questions?

Yes. You can ask clarifying questions which will be shared with the cohort. Questions like “can we mine partial amounts of uranium at a particular location” or “should we separate our report into four sections to align with the criteria?” are fine; questions like “I don’t understand, can you help me come up with some ideas?” are not. Your question should be one which I can publicly provide an answer to.

(Incidentally – yes and yes to those questions.)

Q. What happens if I need more time?

Extensions to the deadline must be agreed in advance, and supported by medical documentation. Otherwise you will receive a zero score for this component. For example:

“Hi, I am off school for the next week with XXXX, here’s a medical certificate.” – fine.

“Hi, I didn’t submit two weeks ago because I was away with a cold, here’s a medical certificate.” – expect zero marks.

Q. Do I get extra marks for using an advanced algorithm?

No. If in doubt, re-read the assessment criteria. Clarity of explanation is more important that a complicated algorithm at this stage.

Q. Can I write my code in Python?

No. Setting up the model would take hours without earning you extra marks. For Part 4 you will be given a model and an algorithm and you will need to write an improved solution to that problem using Python.

Q. I have an ethical objection to the mining of uranium.

No problem - feel free to change the resource being mined to something more palatable to you. Some suggestions are: coal, cobalt, opal, redstone, unicorn feathers, or happy thoughts. Equally, you can change the method of extraction to: gathering, farming, conjuring, or whatever else takes your fancy.

Q. Four hours seems unrealistically short to establish a mine and extract resources.

Yes, I’m surprised too, fortunately it makes the numbers a lot more interesting for the algorithmic problem. It may be that gathering unicorn feathers is more realistic for the numbers given.